Match #2: Windowed-Sinc vs. Chebyshev

Both the windowed-sinc and the Chebyshev filters are designed to separate one band of frequencies from another. The windowed-sinc is an FIR filter implemented by *convolution*, while the Chebyshev is an IIR filter carried out by *recursion*. Which is the best digital filter in the frequency domain? We'll let them fight it out.

The recursive filter contender will be a 0.5% ripple, 6 pole Chebyshev low-pass filter. A fair comparison is complicated by the fact that the Chebyshev's frequency response changes with the cutoff frequency. We will use a cutoff frequency of 0.2, and select the windowed-sinc's filter kernel to be 51 points. This makes both filters have the same 90% to 10% roll-off, as shown in Fig. 21-2(a).

Now the pushing and shoving begins. The recursive filter has a 0.5% ripple in the passband, while the windowed-sinc is flat. However, we could easily set the recursive filter ripple to 0% if needed. No points. Figure 21-2b shows that the windowed-sinc has a much better stopband attenuation than the Chebyshev. One point for the windowed-sinc.

Figure 21-3 shows the step response of the two filters. Both are bad, as you should expect for frequency domain filters. The recursive filter has a nonlinear phase, but this can be corrected with bidirectional filtering. Since both filters are so ugly in this parameter, we will call this a draw.

So far, there isn't much difference between these two filters; either will work when moderate performance is needed. The heavy hitting comes over two critical issues: *maximum performance* and *speed*. The windowed-sinc is a powerhouse, while the Chebyshev is quick and agile. Suppose you have a really tough frequency separation problem, say, needing to isolate a 100

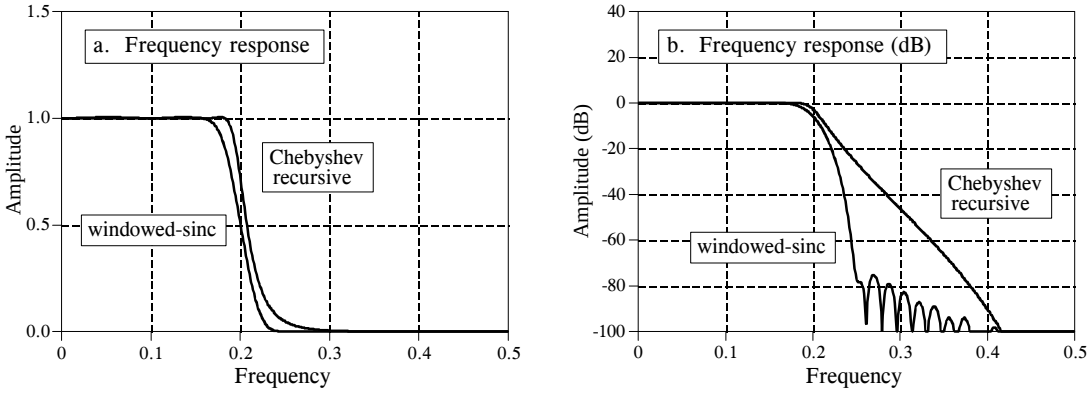

FIGURE 21-2
Windowed-sinc and Chebyshev frequency responses. Frequency responses are shown for a 51 point windowed-sinc filter and a 6 pole, 0.5% ripple Chebyshev recursive filter. The windowed-sinc has better stopband attenuation, but either will work in moderate performance applications. The cutoff frequency of both filters is 0.2, measured at an amplitude of 0.5 for the windowed-sinc, and 0.707 for the recursive.